

Recall: CH II, CH III

To understand "+" and "." = linear operation

vector space - basis - dimension $< \begin{matrix} \infty \\ +\infty \end{matrix}$

Isomorphism Thm = classification

$$\dim_{\mathbb{F}} V = n < +\infty \Leftrightarrow V \cong \mathbb{F}^n \text{ (Need: Isomorphism)}$$

Linear Maps:

FTOL A, $\dim V = n$, $\dim W = m$, then

$$\text{Hom}(V, W) \stackrel{\Phi}{\cong} \mathbb{F}^{m \times n}$$

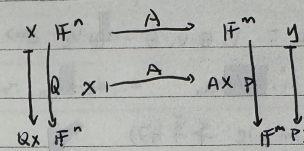
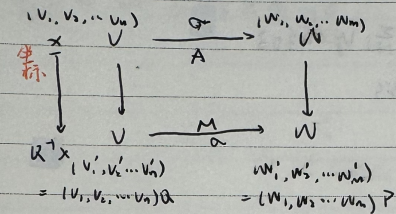
$$\sigma \longmapsto \Phi(\sigma) = A_{\sigma}$$

(i.e. σ sends v with coordinate x to $\sigma(v)$ with coordinate Ax)

Hom (V, W)

$\mathbb{F}^{m \times n}$

$$B = P A Q$$



$$\sigma[(v_1', v_2', \dots, v_n')y] = \sigma[(v_1, v_2, \dots, v_n)(Qy)]$$

$$= (w_1, w_2, \dots, w_m)(A(Qy))$$

$$\sigma[(v_1', v_2', \dots, v_n')y] = (w_1', w_2', \dots, w_m')(My)$$

$$= (w_1, w_2, \dots, w_m)(PM y)$$

$$(w_1, w_2, \dots, w_m)(A(Qy)) = (w_1, w_2, \dots, w_m)(PM y)$$

Conclusion

$A, B \in \mathbb{F}^{m \times n}$ are equivalent $\Leftrightarrow$ A and B are the matrices of the same LM with respect to different bases.

Recall

1. Finite product: $V_1 \times V_2 \times \dots \times V_n \cong (V_1 \times V_2) \times (V_2 \times V_3) \times \dots \times V_n$

$$V \times W = (\{(v, w) \mid v \in V, w \in W\}, +, \cdot)$$

2. $V \times \{0\} \xrightarrow{\tau_1} V \times W \xleftarrow{\tau_2} \{0\} \times W$. [τ_1 = embedding τ_2 = inclusion]

$$(v, 0) \mapsto \tau_1(v, 0) = (v, 0)$$

$$V \times \{0\} \oplus \{0\} \times W = V \times W$$

$$V_i \cong \{0\} \times \{0\} \times \dots \times V_i \times \{0\} \times \dots \times \{0\} = \tilde{V}_i$$

$$V_1 \times V_2 \times \dots \times V_n = \bigoplus_{i=1}^n \tilde{V}_i$$

Conclusion Finite product is equivalent to finite direct sum. Formally,

$$\prod_{i=1}^n V_i \cong \bigoplus_{i=1}^n V_i$$

Recall Finite direct sum in $\mathbb{F}^n$

Let $V_1, V_2, \dots, V_s$ be subspaces of $\mathbb{F}^n$

$$V_1 + V_2 + \dots + V_s = V_1 \oplus V_2 \oplus \dots \oplus V_s \Leftrightarrow V_i \cap \sum_{j \neq i} V_j = \{0\}$$

$$\mathbb{F}^n = \mathbb{F}e_1 \oplus \dots \oplus \mathbb{F}e_n \quad e_1, e_2, \dots, e_n \text{ is a basis}$$

(semi-)simple 半单的

• What about $\bigcap_{i=1}^s V_i$ and $\bigoplus_{i=1}^s V_i$

Any linear map $\bigoplus_{i=1}^s V_i \rightarrow \bigcap_{i=1}^s V_i$ is not an isomorphism

$$i: V \hookrightarrow V \quad (\text{the inclusion map})$$

Ex. ① $\bigoplus_{i=1}^s \mathbb{F} \longrightarrow \bigcap_{i=1}^s \mathbb{F}$ easy

② $\bigoplus_{i=1}^s \mathbb{F} \longrightarrow \bigcap_{i=1}^s \mathbb{F}$ Homework: Find a surjection

Recall. Let $A_1, A_2, \dots, A_n$ be finite sets Find $|A_1 \cup A_2 \cup \dots \cup A_n|$

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$

$$|A_1 \cup A_2 \cup A_3| = \sum |A_i| - \sum_{1 \leq i < j \leq 3} |A_i \cap A_j| + |A_1 \cap A_2 \cap A_3|$$

Thm. ① Let $\dim V_i = d_i < +\infty \quad 1 \leq i \leq n$

$$\dim \bigoplus_{i=1}^n V_i = \sum_{i=1}^n d_i = \dim \bigoplus_{i=1}^n \widetilde{V}_i$$

$$\textcircled{e} \dim (V_1 + V_2) = \dim V_1 + \dim V_2 - \dim (V_1 \cap V_2)$$

$$\text{But } \dim (V_1 + V_2 + V_3) \neq \sum_{i=1}^3 \dim V_i - \sum_{1 \leq i < j \leq 3} \dim (V_i \cap V_j) + \dim (V_1 \cap V_2 \cap V_3)$$

EX. $V_1 = x\text{-axis}$ $V_2 = y\text{-axis}$ $V_3 = \{x=y\}$

$$\text{Left side} = 2 \neq \text{right side} = 3$$

[Four subspace problem]

consider. $\{U \subseteq V \mid U \text{ is a subspace}\} \subseteq 2^V$

Notice $(2^V, \supseteq)$ is a lattice, i.e.

$$\forall x, y \in 2^V \quad x \vee y \text{ and } x \wedge y \text{ exist}$$

$$x \vee y = \sup\{x, y\} \quad x \wedge y = \inf\{x, y\}$$

partial order (偏序)